

SSC Technical Guide: TIS Control to Home Assistant

Full technical reference for reproducing the Phase 4 conversion: bill of materials, software components, configuration, remote access, and the roadmap. Written for residents comfortable with basic networking/Linux concepts. For a plain-language overview, see the [Simple Guide](#); if Home Assistant itself is new to you, start with [What is Home Assistant?](#).

WARNING: Electrical safety - please read first

Parts of this project involve wiring connected to **mains / line voltage electricity**.

- Working with live electrical wiring can cause **serious injury, fire, or death**.
- Any work inside a wall box, distribution board, relay panel, or anything touching mains wiring **must** be performed, or at minimum inspected and signed off, by a qualified electrician familiar with your local electrical code.
- The standard conversion described here needs **no electrical work at all** - the existing switches and sensors stay untouched in their wall boxes and are re-paired over the air. If you ever find yourself about to open a wall box, stop and call an electrician.

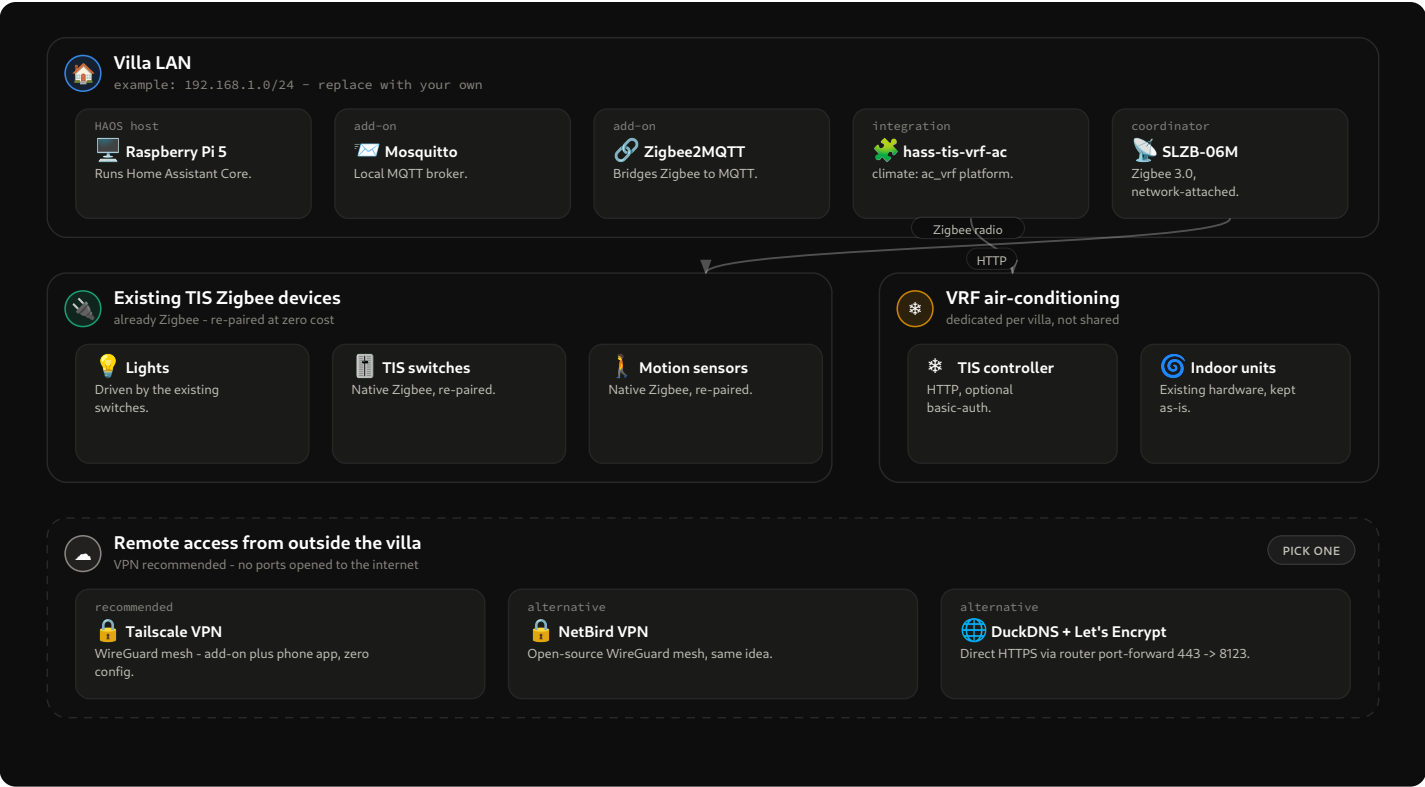
IMPORTANT: Disclaimer - no warranty, use at your own risk

This documentation and the linked custom integration ([hass-tis-vrf-ac](#)) are provided "**as is**" as a community resource, with **no warranty of any kind** and **no guarantee of fitness for any particular purpose**. The author and contributors are **not liable** for any damage, injury, death, fire, data loss, loss of climate control, security incident, or other loss arising from using this material. This is a personal DIY project inside your own villa - community management is not involved and provides no support; you are responsible for complying with local law, electrical code, and any agreements that apply to your villa. Every system touched here (switches, sensors, VRF air-conditioning) is dedicated to your own villa and not shared with neighbours. All IP addresses, domains, usernames, and passwords shown in this guide are **placeholders** - replace them with your own values and never commit real credentials to a public repository.

Table of Contents

1. Architecture
2. Bill of materials
3. Installing Home Assistant OS on the Raspberry Pi 5
4. Setting up Zigbee2MQTT with the SLZB-06M
5. Re-pairing the existing TIS switches and motion sensors
6. Controlling the villa's VRF air-conditioning
7. Remote access
 - 7.1. Option A (recommended): Tailscale
 - 7.2. Option B: DuckDNS + Let's Encrypt (direct exposure)
8. Backups and updates
9. Automations
10. Troubleshooting
11. Roadmap: RS485 sensor bus integration
12. Contributing back

1. Architecture



Components:

Component	Role
Raspberry Pi 5, running Home Assistant OS (HAOS)	Host for everything below. Runs Home Assistant Core plus add-ons in isolated containers.
Home Assistant Core	Automations, dashboards, users/auth, and remote-access endpoint.
Mosquitto add-on	Local MQTT broker. Zigbee2MQTT publishes device state here; Home Assistant subscribes to it via the official MQTT integration.
Zigbee2MQTT add-on	Talks to the Zigbee coordinator, exposes every paired Zigbee device over MQTT.
SMLIGHT SLZB-06M	Zigbee 3.0 coordinator. Unlike a plain USB dongle, it's network-attached (Ethernet or Wi-Fi, with PoE support), so Zigbee2MQTT connects to it over TCP instead of needing USB passthrough into the HAOS VM - simpler and more reliable on a Raspberry Pi host.
Existing TIS Zigbee devices	The light switches and motion sensors TIS installed are native Zigbee devices. They are re-paired from the TIS network to your own coordinator - no new hardware, no rewiring, no cost.
hass-tis-vrf-ac custom integration	Community-written Home Assistant <code>climate</code> platform (source) that talks HTTP directly to the villa's own TIS VRF air-conditioning controller, so the original AC hardware keeps working under Home Assistant instead of the TIS Control panel.
TIS VRF AC controller	Existing proprietary hardware, kept in place. Dedicated to this villa only - each villa has its own controller and AC unit, not a system shared across the community. Reachable on the villa's LAN over HTTP, optionally with HTTP Basic Auth.

Component	Role
Remote access (Tailscale, or alternatives)	Optional secure path to reach Home Assistant from outside the villa - see Remote access .

Everything except the optional remote-access path works with **no internet connection**.

2. Bill of materials

The switches, motion sensors, and wiring that TIS Control used to drive are not on this list - they are already in your villa, they are already Zigbee, and they get re-paired rather than replaced. **The re-use is free**. What you buy is only the new brain and its radio:

Item	Notes	Approx. role
Raspberry Pi 5, 8 GB RAM (64-bit)	The HAOS host. The 8 GB model gives comfortable headroom for add-ons; use the official 27 W USB-C power supply.	The brain
microSD card, 64 GB or larger	A2-rated card from a reputable brand. (An SSD via USB is a worthwhile future upgrade for longevity, but a good SD card is fine to start.)	Storage
SMLIGHT SLZB-06M	Zigbee 3.0 coordinator with Ethernet/Wi-Fi/USB and PoE options. Connect it to your LAN via Ethernet rather than USB - it runs independently of the Pi.	Zigbee radio
Ethernet cable(s)	For the Pi and (recommended) the SLZB-06M. Wired beats Wi-Fi for reliability.	Network

That's the whole list - roughly the cost of a mid-range phone accessory haul, with no recurring fees.

3. Installing Home Assistant OS on the Raspberry Pi 5

1. Download the official Raspberry Pi 5 HAOS image from the [Home Assistant installation page](#).
2. Flash it to your SD card using [Raspberry Pi Imager](#) or balenaEtcher .
3. Boot the Pi with it connected to your router via Ethernet.
4. Browse to `http://homeassistant.local:8123` from a device on the same network and complete the onboarding wizard (create your admin account - use a strong, unique password).
5. Apply any pending HAOS/Core updates from **Settings** → **System** → **Updates** before continuing.

4. Setting up Zigbee2MQTT with the SLZB-06M

1. Connect the SLZB-06M to your network (Ethernet recommended) and give it a fixed IP, either via a DHCP reservation on your router or a static IP on the device itself. This guide uses `192.0.2.50` as a placeholder - use your own address.
2. Install the **Mosquitto broker** add-on from **Settings** → **Add-ons** → **Add-on Store**, start it, and create a dedicated MQTT user for Zigbee2MQTT (do not reuse your HA admin account).

3. Install the **Zigbee2MQTT** add-on from the add-on store.
4. Configure the add-on (via its **Configuration** tab, or `configuration.yaml` inside the add-on) to point at the coordinator over the network rather than USB:

```
serial:
  port: tcp://192.0.2.50:6638 # <-- replace with your SLZB-06M's own IP
  adapter: ezsp

mqtt:
  base_topic: zigbee2mqtt
  server: mqtt://core-mosquitto:1883
  user: <MQTT_USERNAME>
  password: <MQTT_PASSWORD_REDACTED>
```

5. Start the add-on and open its web UI (via the **Zigbee2MQTT** panel in the HA sidebar) to confirm the coordinator connects.
6. In Home Assistant, add the **MQTT** integration (**Settings** → **Devices & Services** → **Add Integration** → **MQTT**) pointing at the same broker - Zigbee2MQTT devices will then appear automatically via MQTT discovery.

5. Re-pairing the existing TIS switches and motion sensors

This is the heart of the zero-cost story: the TIS wall switches and motion sensors are standard Zigbee devices currently paired to the TIS network. Moving them to yours is a software operation done from the room, not the wall box.

1. In the Zigbee2MQTT web UI, enable **Permit join**.
2. Put the device into pairing mode. The exact button-press or power-cycle sequence varies by device model - ask in the WhatsApp group; the neighbours who've converted already know the sequence for each TIS device type in our villas, and this section will grow as each model is documented.
3. Watch the Zigbee2MQTT log: the device joins, gets interviewed, and appears with its capabilities.
4. Rename it immediately with a clear, consistent scheme (e.g. `living_room_ceiling_light`, `hallway_motion`) - this pays off enormously when writing automations later.
5. Repeat, one room at a time. There's no rush, and devices still paired to TIS keep working under TIS until you move them.

Keep the coordinator away from Wi-Fi routers/USB 3.0 devices where possible; 2.4 GHz interference is the most common cause of a flaky Zigbee network.

6. Controlling the villa's VRF air-conditioning

The `hass-tis-vrf-ac` integration is a custom `climate` platform that talks HTTP directly to the TIS VRF AC controller that was already on your network, so the original AC hardware keeps working under Home Assistant. Each villa has its own controller and AC unit - this touches nothing outside your own home.

1. Confirm the TIS VRF AC controller's IP address on your LAN (check your router's client list, or the original TIS Control installation documentation).
2. Install the integration, following the [repository's own INSTALLATION.md](#) - either via HACS (custom repository) or by copying `custom_components/ac_vrf` into your HAOS `config/custom_components` folder (accessible via the **Samba** or **File editor** add-on, or SSH).
3. Restart Home Assistant, then add the platform to your configuration. For example, in `configuration.yaml`:

```

climate:
  - platform: ac_vrf
    host: 192.0.2.251          # <-- replace with your AC controller's IP
    username: <AC_CONTROLLER_USERNAME> # optional, only if your controller requires auth
    password: <AC_CONTROLLER_PASSWORD_REDACTED> # optional, can be blank if unset

logger:
  default: info
  logs:
    custom_components.ac_vrf: debug      # helpful while getting this working, remove after

```

- Restart Home Assistant again and confirm a new climate entity appears for each AC zone/unit. Test basic on/off and temperature-setpoint changes before wiring it into any automations.
- Check the repository's own [TROUBLESHOOTING.md](#) and the included connection-test script if the controller doesn't respond - it's usually an IP, firewall, or auth mismatch rather than a Home Assistant problem.

7. Remote access

Controlling your villa from outside your home Wi-Fi is genuinely useful (AC on before you arrive, checking things while travelling), so this section is part of the standard build rather than an afterthought. There are two good free approaches; **the VPN approach is recommended** because it exposes nothing to the internet.

	VPN mesh (recommended)	DuckDNS + Let's Encrypt
How it works	Your phone and the Pi join a private encrypted network (WireGuard-based); you reach HA as if you were at home.	Your router forwards port 443 to HA, protected by TLS and your HA login.
Exposure to internet	None - no open ports.	One HTTPS port open to the whole internet.
Setup difficulty	Easy - install two apps, log in.	Moderate - DNS, certificates, router port-forwarding.
Works behind CG-NAT	Yes.	No (needs a real public IP).
Cost	Free tier is ample for a household.	Free.

7.1. Option A (recommended): Tailscale

[Tailscale](#) is a zero-configuration WireGuard mesh - the easiest route by far.

- Install the **Tailscale** add-on from the Home Assistant add-on store and start it.
- Open the add-on's Web UI and log in with a free Tailscale account (Google/ Microsoft/GitHub sign-in supported).
- Install the Tailscale app on your phone and log in to the same account.
- That's it - from anywhere, open `http://<the-pi's-tailscale-name>:8123` (or its `100.x.y.z` Tailscale IP) in the HA companion app.

[NetBird](#) is an equally good open-source alternative with a similar flow (add-on/agent on the Pi, app on the phone, same account on both). If you prefer fully self-hosted infrastructure or hit a Tailscale limitation, use it the same way.

7.2. Option B: DuckDNS + Let's Encrypt (direct exposure)

The classic free path documented by Home Assistant, for those who want a plain `https://` URL that works from any browser without a VPN app.

This forwards a port from the internet to your Pi. Only choose it if you understand the trade-off, and follow the hardening steps below.

1. Register a free hostname at duckdns.org, e.g. `<your-subdomain>.duckdns.org`, and note the DuckDNS token (treat it like a password).
2. Install the official **DuckDNS** add-on and configure it:

```
lets_encrypt:
  accept_terms: true
  certfile: fullchain.pem
  keyfile: privkey.pem
  token: <DUCKDNS_TOKEN_REDACTED>
  domains:
    - <your-subdomain>.duckdns.org
  aliases: []
```

yaml

3. Point Home Assistant at the certificate in `configuration.yaml`:

```
http:
  ssl_certificate: /ssl/fullchain.pem
  ssl_key: /ssl/privkey.pem
```

yaml

4. Forward external port `443` to the Pi's port `8123` on your router.
5. Set **Settings** → **System** → **Network** → **Home Assistant URL** to `https://<your-subdomain>.duckdns.org`, restart, and test from mobile data.

Hardening (mandatory if you choose Option B):

- Strong, unique admin password and [two-factor authentication](#) on every account.
- A separate non-admin account for day-to-day household use.
- Enable the failed-login IP ban threshold in the `http:` configuration.
- Keep HAOS, Core, and add-ons updated.
- If you later switch to Tailscale, remove the port forward and disable the DuckDNS add-on.

8. Backups and updates

- Enable Home Assistant's built-in [automatic backups](#) (**Settings** → **System** → **Backups**) before making significant changes, and before every major HAOS/Core update.
- Store at least one backup copy off the Pi itself (network share, cloud storage, USB drive) so an SD-card failure doesn't take your only backup with it.
- Before re-pairing devices in bulk, take a fresh backup - the Zigbee network state lives in Zigbee2MQTT's data and is included in a full HA backup.

9. Automations

Once devices are paired, the fun starts. Rather than repeating examples here, the community keeps a living library in the repository:

- [automations/common/](#) - works in any villa
- [automations/phase-4/](#) - specific to the TIS switch/motion-sensor/VRF-AC hardware in Phase 4

Copy, adapt, and contribute back - see [the contribution notes](#).

10. Troubleshooting

Symptom	Likely cause / where to look
Zigbee2MQTT add-on won't connect to the coordinator	Wrong IP/port for the SLZB-06M, or it's not reachable on the network. Check its own web UI (reachable directly by browsing to its IP) and the add-on logs.
A TIS device won't pair	It may still be joined to the TIS network - it must be removed/reset first (pairing-mode sequence varies by model; ask in the WhatsApp group). Also check Permit join is actually enabled and the device is within range.
Zigbee devices randomly drop offline	Usually 2.4 GHz interference (Wi-Fi routers, USB 3.0 cables/hubs) or the coordinator being too far from devices. Try relocating the coordinator.
ac_vrf climate entity doesn't appear	Check Settings → System → Logs for <code>custom_components.ac_vrf</code> errors, verify the controller's IP is correct and reachable (ping), and see the integration's own TROUBLESHOOTING.md .
Can't reach HA remotely via Tailscale	Confirm both the add-on and the phone app are logged in to the same Tailscale account, and the add-on shows a <code>100.x.y.z</code> address in its log.
Can't reach HA remotely via DuckDNS	Confirm the port forward is active (online port-checker from outside), the DuckDNS add-on log shows successful cert issuance, and your ISP isn't using carrier-grade NAT (which blocks inbound port forwarding - use Tailscale instead in that case).

11. Roadmap: RS485 sensor bus integration

The villas carry an **RS485 bus** connecting additional built-in hardware that the TIS panel used but the current conversion doesn't reach yet:

- additional **motion sensors** beyond the Zigbee ones
- **water-usage** metering
- **power-usage** metering

The plan is to bridge this bus into Home Assistant (typically via an RS485-to- Ethernet/USB gateway plus a Modbus or custom integration, depending on what protocol analysis reveals). This is under investigation in the villa that did the first conversion - findings, wiring notes, and configuration will be added to this guide and the automation library as they solidify. If you have RS485 or protocol-reversing experience, this is the most valuable place to contribute.

12. Contributing back

If you improve on this setup - a clearer config, a documented pairing sequence for a TIS device model, an RS485 discovery, an automation worth sharing - please contribute it back (pull request, or pass it to whoever maintains this repository) so the next neighbour benefits too.